WEEK 2 · DAY 2 · OVERVIEW

Volatility Clustering and GARCH



Created by **Jeff Lenamon**

ABOUT 55 MIN WITH EXERCISES / 11 SECTIONS / 5 EXERCISES

This course is educational content created in a personal capacity. Nothing here is investment advice or a recommendation to buy or sell any security. All portfolio data is synthetic.

Problem Statement

BUSINESS CONTEXT

Days 4 and 6 found memory in the size of a daily change: its square remembers the days before, and an EWMA of the size follows it.

A risk desk wants one model of that size it can read: how stormy a date was, from earlier dates only, and how long a storm takes to fade.

Today fits GARCH(1,1) to the five series' daily changes with arch, reads each of its numbers, checks its shocks, and scores it walk-forward beside Day 6's EWMA.

A fitted volatility describes the file's past dates; it is never a risk call and never a forecast.

OBJECTIVES

Measure how a series' largest days sit in time, write the GARCH(1,1) variance recursion by hand and match arch, and read omega, alpha, beta, persistence, half-life and long-run volatility.

Compare three measures of volatility, choose the shock curve and the variance model by AIC and BIC, score a one-step volatility walk-forward on the size of a change, and set the size beside the level.

ABOUT 55 MIN WITH EXERCISES / 11 SECTIONS / 5 EXERCISES

Your Route Through Day 2

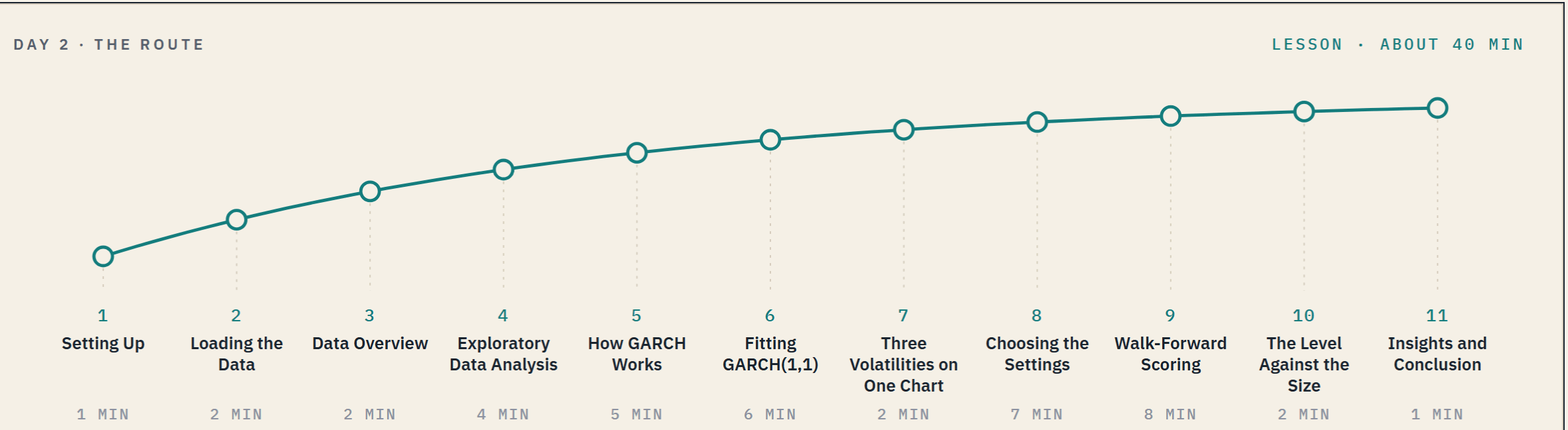


FIGURE 1 Eleven sections, in order, with about how long each takes.

Time. About 40 minutes for the lesson, then about 15 for the five exercises.

Fix the seed and load the libraries.

Setting Up

THE SETUP

```
import os
from pathlib import Path
SEED = 42
DATA = "../../../data/"
if not Path(DATA + "benchmark.csv").exists():
    DATA = "https://raw.githubusercontent.com/lenamonj/python-fixed-income/main/data/"
```

THE LIBRARIES

```
import numpy as np
import pandas as pd
from scipy import stats
from sklearn.model_selection import TimeSeriesSplit
from sklearn.metrics import mean_absolute_error
from statsmodels.tsa.stattools import acf
from statsmodels.tsa.arima.model import ARIMA
from arch import arch_model
```

Loading the Data

READING THE TWO FILES

```
yields = pd.read_csv(DATA + "treasury_par_yields.csv", parse_dates=["date"], index_col="date")
spreads = pd.read_csv(DATA + "course3_synthetic_spreads.csv", parse_dates=["date"], index_col="date")
CUT = "2022-12-30"
development = yields.loc[:CUT]
```

FIVE SERIES IN BP, AND THEIR CHANGES

```
levels = pd.DataFrame({"2 Yr": (development["2 Yr"] * 100).round(2),
                       "10 Yr": (development["10 Yr"] * 100).round(2),
                       "2s10s": ((development["10 Yr"] - development["2 Yr"]) * 100).round(2),
                       "IG": spreads.loc[:CUT, "ig_spread_bp"],
                       "HY": spreads.loc[:CUT, "hy_spread_bp"]})
changes = levels.diff().dropna()
print(len(levels), "levels and", len(changes), "changes, from", changes.index[0].date(), "to", changes.index[-1].date())
```

```
2001 levels and 2000 changes, from 2015-01-05 to 2022-12-30
```

Data Overview

THE FIVE SERIES' CHANGES, IN BP

	2 YR	10 YR	2S10S	IG	HY
count	2000.00	2000.00	2000.00	2000.00	2000.00
mean	0.19	0.09	-0.10	0.02	0.05
std	4.19	5.03	3.32	1.19	5.01
min	-27.00	-30.00	-16.00	-7.00	-23.00
25%	-2.00	-3.00	-2.00	-0.70	-2.80
50%	0.00	0.00	0.00	0.00	0.00
75%	2.00	3.00	2.00	0.60	2.80
max	34.00	29.00	18.00	9.80	42.90

EXCESS KURTOSIS

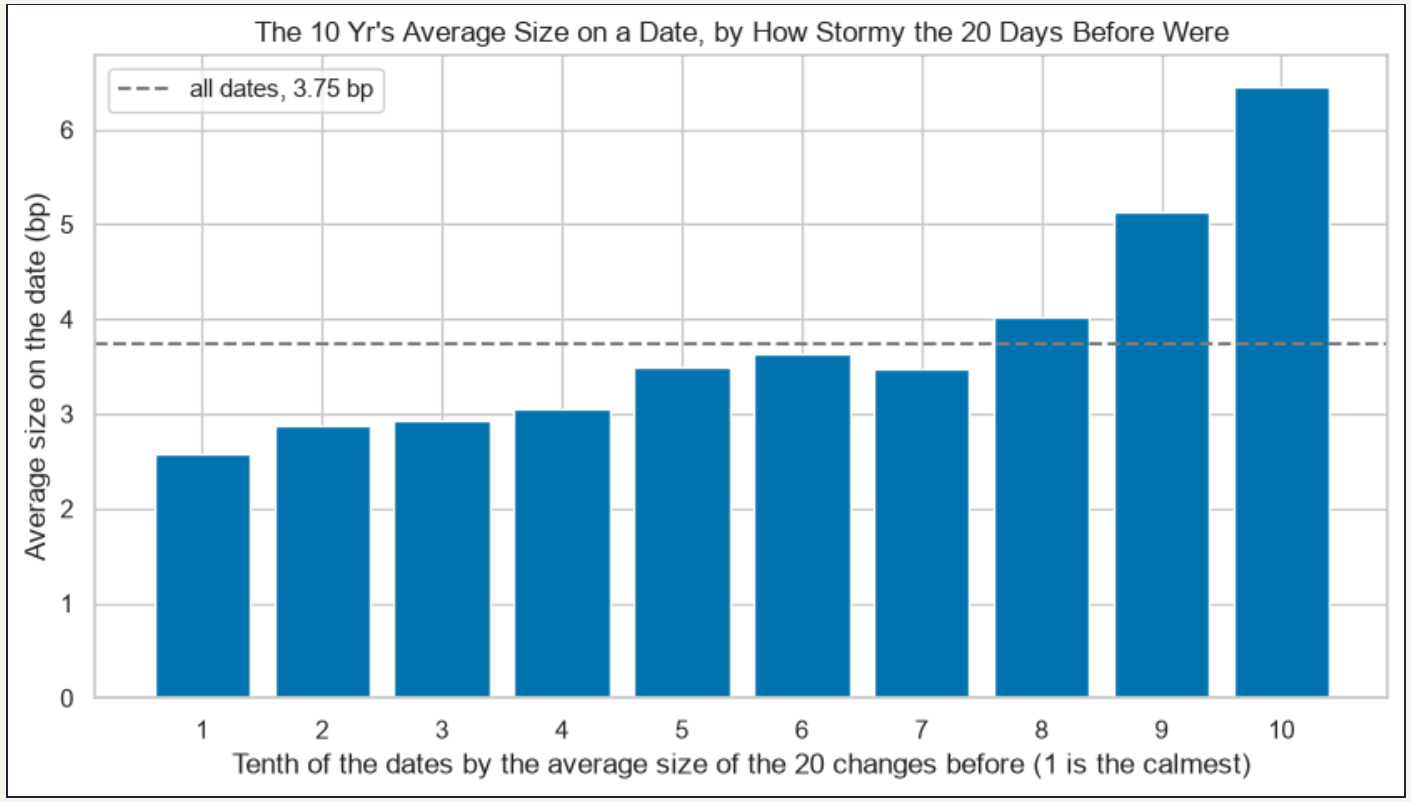
```
changes.apply(stats.kurtosis).round(2)

2 Yr     7.68
10 Yr    2.83
2s10s    2.13
IG       7.14
HY       6.33
dtype: float64
```

Too many large days for one bell curve; next, whether they scatter or sit together.

Find where the largest days fall in time.

Exploratory Data Analysis



TEN GROUPS BY THE 20 DAYS BEFORE

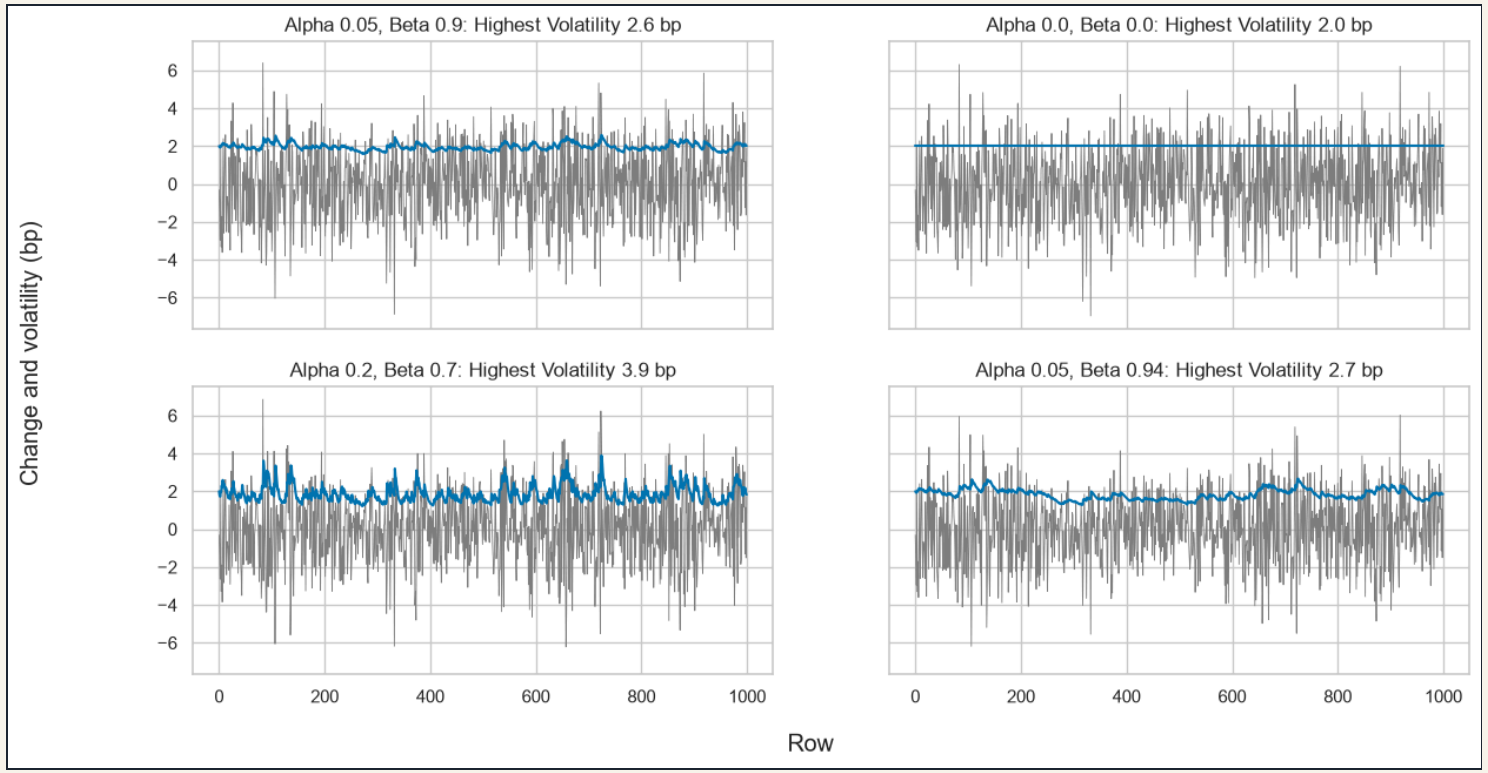
```
before = size.shift(1).rolling(20).mean()
tenth = pd.qcut(before, 10, labels=range(1, 11))
by_tenth = size.groupby(tenth, observed=True).mean()
```

AUTOCORRELATION AT LAG 1

```
Change: -0.007 | size: 0.211 | square: 0.225
```

The sign forgets; the size remembers. A model of the size needs a variance that carries yesterday into today.

How GARCH Works



THE FLOOR

omega: a small floor, the same on every row.

THE WEIGHT ON THE SHOCK

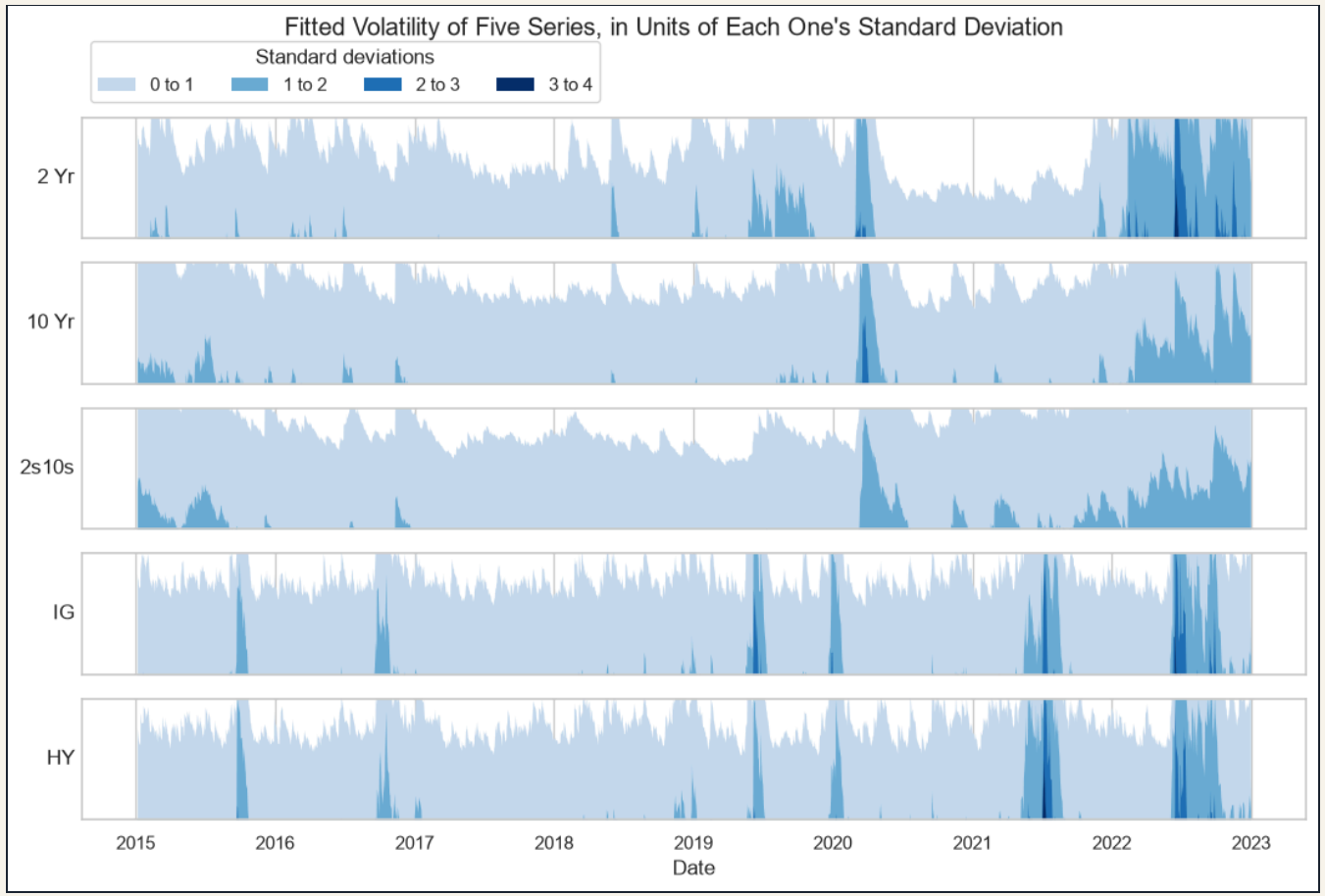
alpha: the weight on yesterday's squared shock.

THE WEIGHT ON THE VARIANCE

beta: the weight on yesterday's variance.

Alpha says how hard a shock hits; persistence says how long it stays.

Fitting GARCH(1,1)



THE 10 YR'S FIT

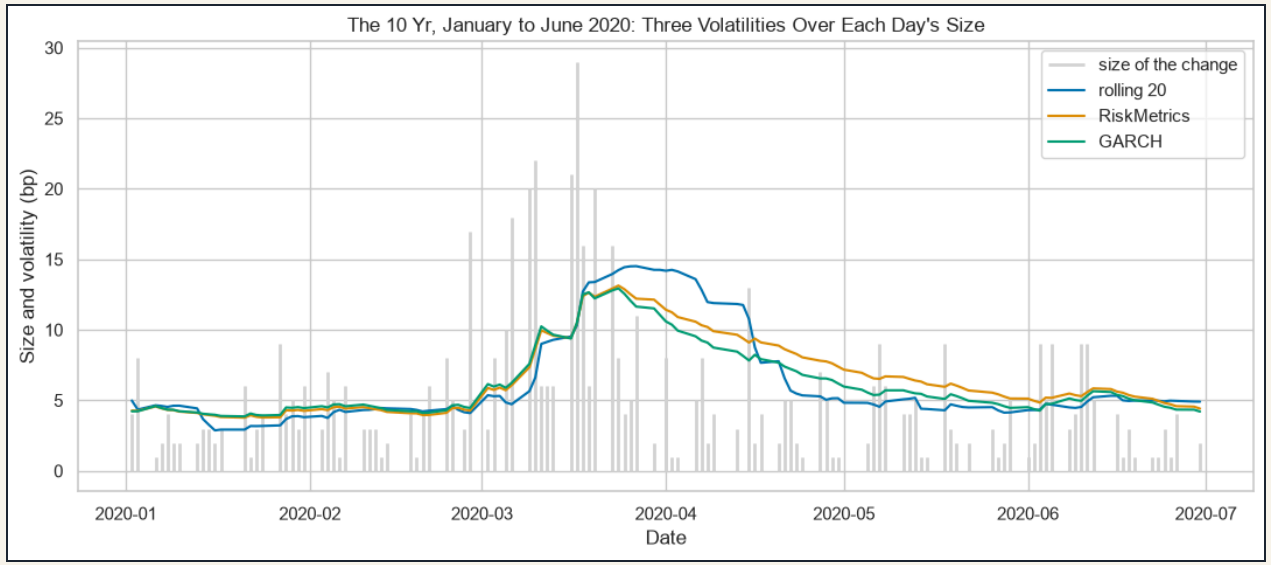
	ESTIMATE	STANDARD ERROR
mu	0.0715	0.0929
omega	0.4668	0.2421
alpha[1]	0.0658	0.0171
beta[1]	0.9151	0.0248

THE RECURSION BY HAND

```
Equal to arch's: True | largest gap: 0.0 bp
```

Five lines give arch's 2,000 volatilities to the last digit.

Three Volatilities on One Chart



HOW ALIKE THE THREE ARE

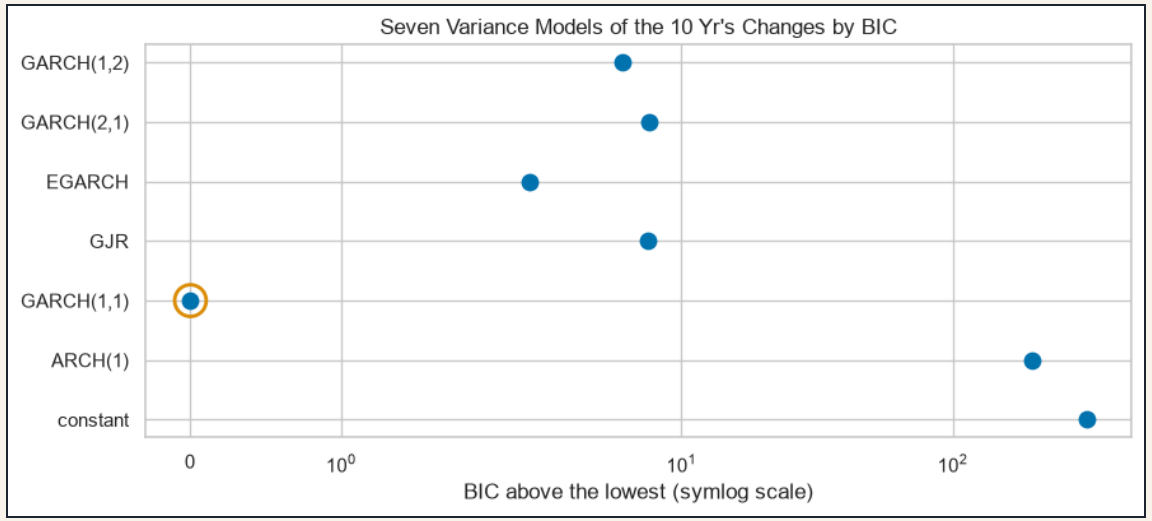
	rolling 20	RiskMetrics	GARCH
rolling 20	1.000	0.959	0.961
RiskMetrics	0.959	1.000	0.987
GARCH	0.961	0.987	1.000

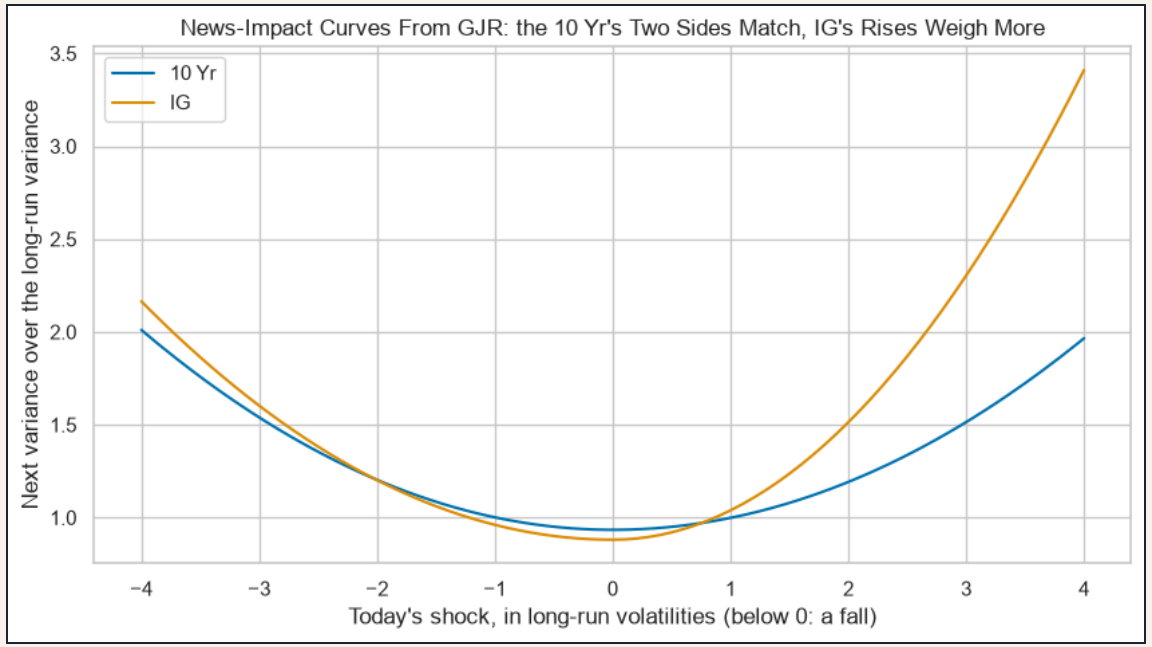
THREE VOLATILITIES, EACH FROM THE ROWS BEFORE IT

```
three = pd.DataFrame({"rolling 20": ten.rolling(20).std().shift(1),
                      "RiskMetrics": np.sqrt((ten ** 2).ewm(alpha=0.06, adjust=False).mean().shift(1)),
                      "GARCH": volatility})
```

Three measures, one storm; GARCH chooses its fading from the data.

Choosing the Settings

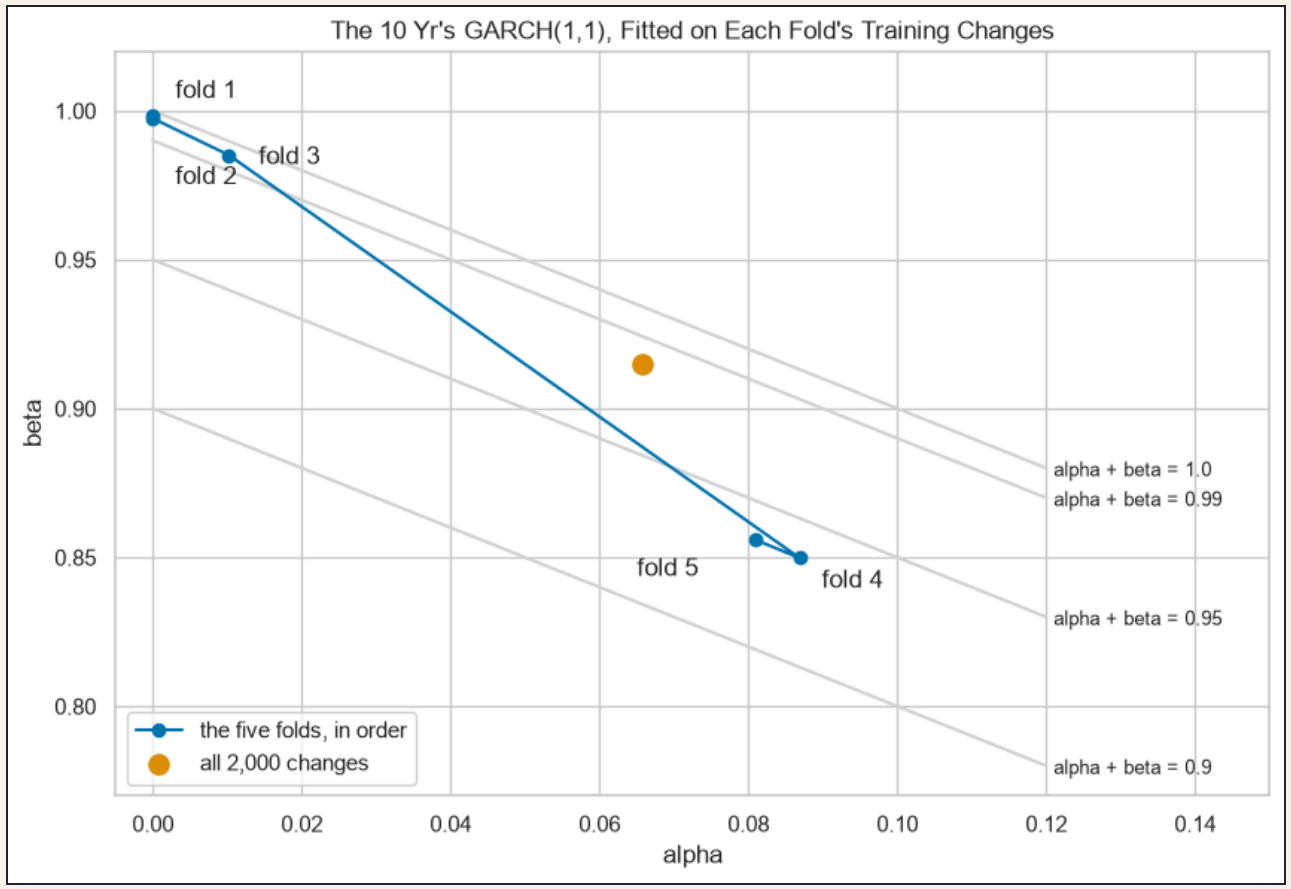


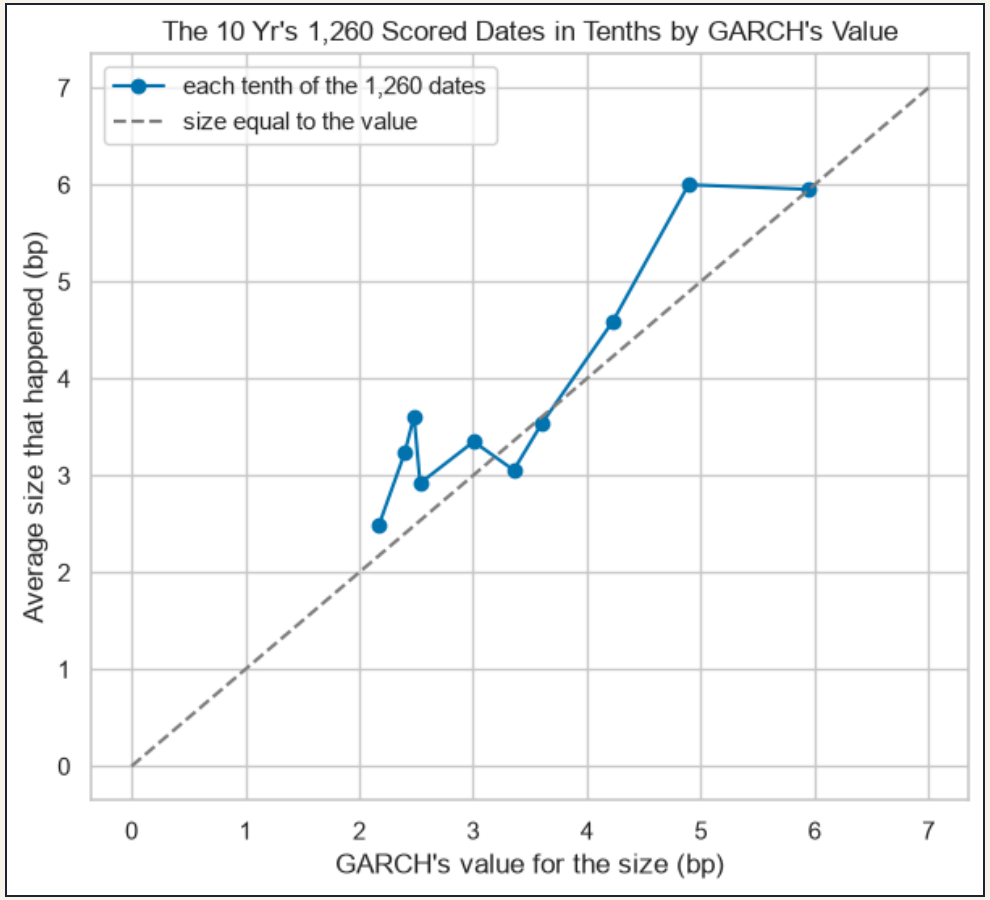


GAMMA, WITH AND WITHOUT

```
10 Yr | gamma: 0.003 | BIC with gamma: 11846.1 | without: 11838.6
IG | gamma: -0.078 | BIC with gamma: 5814.1 | without: 5814.6
```

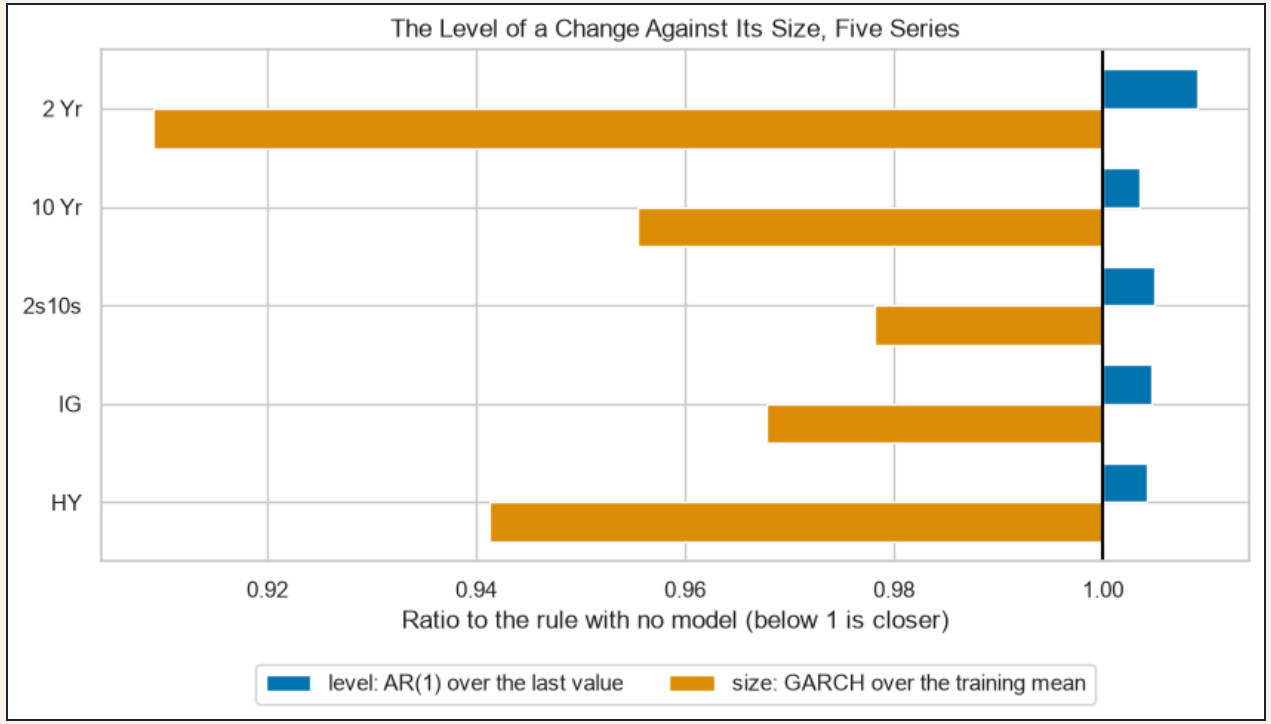
Walk-Forward Scoring





The calmest tenth averaged 2.48 bp and the stormiest 5.95

The Level Against the Size

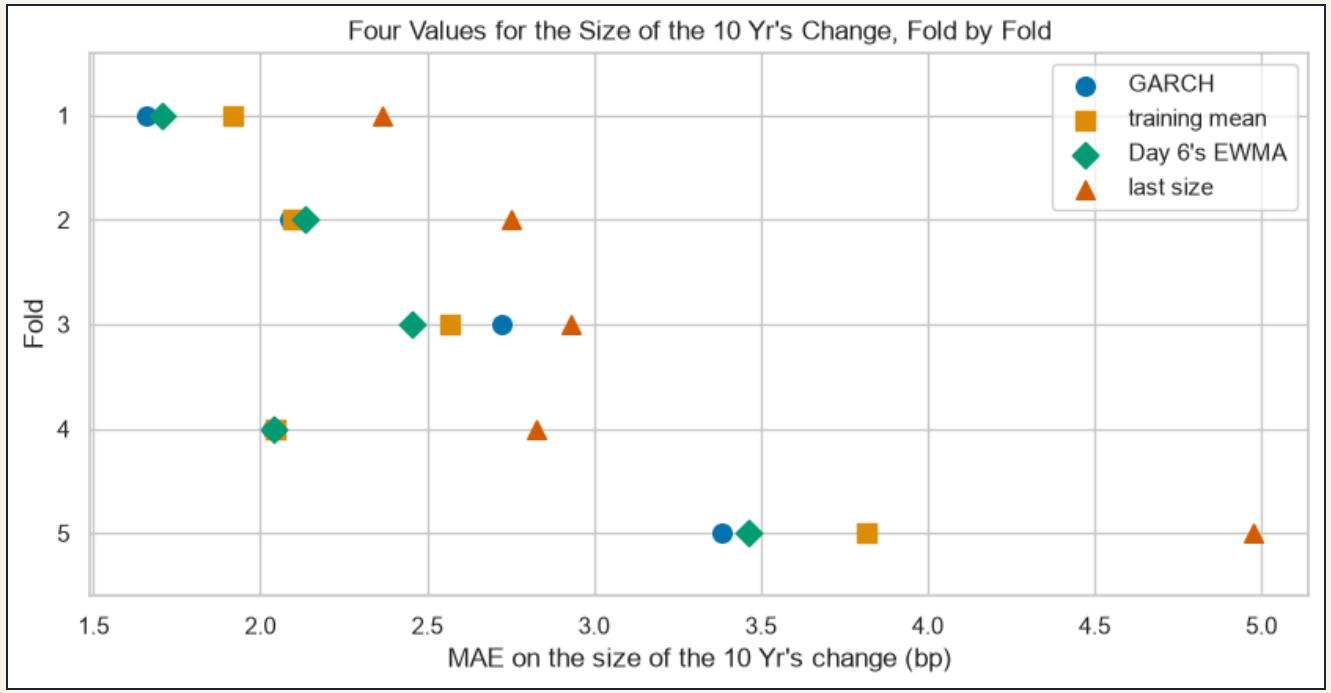


GARCH'S SIZE MAE AGAINST TWO RULES

SERIES	GARCH / MEAN	GARCH / EWMA	FOLDS CLOSER
2 Yr	0.909	1.023	5
10 Yr	0.955	1.008	4
2s10s	0.978	0.998	3
IG	0.968	0.976	4
HY	0.941	0.975	5

On the level of a change, the AR(1) is no closer than the last value; on its size, GARCH is closer than a constant on every series, by less than the folds disagree.

Insights and Conclusion



MAE ON THE SIZE, IN BP

FOLD	GARCH	TRAINING MEAN	DAY 6'S EWMA
1	1.657	1.916	1.706
2	2.089	2.098	2.134
3	2.723	2.566	2.454
4	2.036	2.047	2.041
5	3.382	3.814	3.462
average	2.377	2.488	2.359

Model the size, carry the level.

You Can Now

- Measure whether a series' largest days scatter or cluster in time.
- Write the GARCH(1,1) recursion by hand and match arch's fitted volatility to the last digit.
- Fit **`arch_model`** and read omega, alpha, beta, persistence, half-life and long-run volatility.
- Choose the shock curve and the variance model by AIC and BIC, and read a news-impact curve.
- Score a volatility walk-forward on the size of a change, and set the size beside the level.

NEXT
WEEK 2 · DAY 3

Lag Features and Gradient Boosting

Day 8 hands LightGBM many inputs about the past, and the cut test catches a leak.